

Stock and Flow 101 **Guiding Problem Book and Master Reference**

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Finite balances first; rates second; limits only when needed.

How to use this guide. Choose the system, identify the stock, identify the finite flow, write the finite balance, and only then introduce a rate if it is useful.

I. Defining Diagram and Standard Translation

The basic notation is

$$S_i \xrightarrow{\Downarrow \mathcal{F}} S_f$$

where S_i and S_f are the initial and final stocks, respectively, and $\mathcal{F}$ is the net inflow accumulated during the transition.

The diagram is translated algebraically as

$$S_i + \mathcal{F} = S_f, \quad \text{or equivalently} \quad \boxed{\mathcal{F} = S_f - S_i.}$$

Thus the net inflow $\mathcal{F}$ is determined by the initial and final stocks. We call $S_f - S_i$ the final-minus-initial difference, or simply the *gain* in S .

If the two states are also labeled by numerical times $t_i < t_f$, set

$$\Delta t = t_f - t_i > 0.$$

Define the average net flow rate $\bar{I}$ by

$$\bar{I}_{[t_i, t_f]} = \frac{\mathcal{F}}{\Delta t} = \frac{S_f - S_i}{t_f - t_i}.$$

Hence

$$\boxed{S_f - S_i = \bar{I} \Delta t.}$$

Three levels

S stock,

$\mathcal{F} = S_f - S_i$ gain,

$\bar{I} = \mathcal{F}/\Delta t$ average flow rate.

Units:

$$[\mathcal{F}] = [S], \quad [\bar{I}] = [S]/[t].$$

Finance/accounting examples

Question 1. A checking account begins with \$1,200. During the next 14 days, \$1,800 enters and \$1,200 leaves. Find the final balance, the gain, and the average net cash-flow rate.

Answer. Let the balance be the stock B . The net finite inflow is

$$\mathcal{F} = \$1800 - \$1200 = \$600.$$

Namely

$$B_i \xrightarrow[\Downarrow \$1200]{\Downarrow \$1800} B_f$$

Thus $B_i \xrightarrow{\Downarrow \$600} B_f$:

$$B_f = B_i + \mathcal{F} = \$1200 + \$600 = \$1800,$$

and

$$\bar{I} = \frac{\$600}{14 \text{ days}} \approx \$42.86/\text{day}.$$

This does not mean that \$42.86 entered every day.

Question 2. Treat outstanding debt D as a stock. During one billing period there is new borrowing, interest/fees, and a payment. Write the finite stock-flow balance.

Answer.

$$D_f - D_i = (\text{new borrowing}) + (\text{interest/fees}) - (\text{payments}).$$

The finite bookkeeping is exact before any compounding model is introduced:

$$D_i \xrightarrow[\Downarrow \text{payments}]{\Downarrow \text{net borrowing plus interest/fees}} D_f$$

Physical realizations at a glance

Context	Stock / finite flow	Average rate
Momentum	p , impulse J	force F_{av}
Energy	E or K , work/heat	power P_{av}
Cash	balance B , net cash flow	cash-flow rate
Inventory	amount N , net units	throughput

Exact finite logic

Conservation over the chosen interval means zero net finite flow:

$$\boxed{\mathcal{F} = 0 \iff S_f = S_i.}$$

If the flow rate is constant with value I_0 , then its average rate is also $\bar{I} = I_0$, and

$$S_f - S_i = I_0 \Delta t.$$

The sign of I_0 tells whether the stock has net increased, decreased, or remained unchanged.

Question 3. A savings account begins at \$4,000. During a 30-day month, \$1,500 is deposited, \$900 is withdrawn, and \$20 is credited as interest. Find the final balance, gain, and average net flow rate.

Answer.

$$\mathcal{F} = \$1500 - \$900 + \$20 = \$620,$$

$$B_f = \$4000 + \$620 = \$4620,$$

$$\bar{I} = \frac{\$620}{30 \text{ days}} \approx \$20.67/\text{day}.$$

Again, the average rate summarizes the endpoint gain; it need not describe any particular day.

Formulation checklist

Before calculating, declare

$$\boxed{\text{system} - \text{stock} - \text{flow} - \text{interval} - \text{sign}.$$

II. General Framework Questions

A1. Diagram. Draw the basic stock-flow diagram and translate it into an equation: $S_i \xrightarrow{\mathcal{F}} S_f$.

A2. Gain. Starting from $S_i + \mathcal{F} = S_f$, solve for $\mathcal{F}$ and identify the gain in S : $\mathcal{F} = S_f - S_i$.

A3. Average rate. If time labels are attached to the two states, define $\bar{I} = \mathcal{F}/\Delta t$.

A4. Conservation. Show that

$$\mathcal{F} = 0 \iff S_f = S_i.$$

Interpret the result.

A5. Sign. Explain the meanings of

$$\mathcal{F} > 0 \quad (\text{gain}), \quad \mathcal{F} < 0 \quad (\text{loss}), \quad \mathcal{F} = 0 \quad (\text{balanced}).$$

A6. Stock, flow, rate. Explain the conceptual and dimensional differences among

$$S, \quad \mathcal{F} = S_f - S_i, \quad \bar{I} = \mathcal{F}/\Delta t.$$

A7. Mechanics realization. Give a Newtonian example and identify stock, finite flow, and average rate.

A8. Finance realization. Give a financial example and identify initial stock, final stock, finite flow, and average flow rate.

Question 4. A tank contains 120 L of water. During an hour, 50 L enters and 30 L leaves. Treat the water amount as the stock. Find the final stock, gain, and average net flow rate.

Answer.

$$\begin{aligned} 120 \text{ L} &\xrightarrow{50 \text{ L} - 30 \text{ L}} S_f \\ S_f &= 120 \text{ L} + 20 \text{ L} = 140 \text{ L}, \\ \bar{I} &= 20 \text{ L h}^{-1}. \end{aligned}$$

IV. ★ Calculus Bridge

Presentation note. This section only shows how the finite formulation connects to conventional calculus notation. Unless an instructor explicitly requires it, the finite stock-flow law is the main formulation.

If the local limit exists,

$$I(t) = \lim_{\Delta t \rightarrow 0} \frac{S(t + \Delta t) - S(t)}{\Delta t},$$

so

$$\frac{dS}{dt}(t) = I(t).$$

The differential law is a limiting refinement of the finite balance.

C1. Three levels. Classify

$$S_f - S_i = \mathcal{F}, \quad S_f - S_i = \bar{I}\Delta t, \quad \frac{dS}{dt} = I.$$

Which one requires a limit?

C2. Units. If $[S] = Q$, show that both $\bar{I}$ and dS/dt have units Q/time .

V. Momentum as a Stock

For a chosen object or system,

$$\mathbf{p} = m\mathbf{v}.$$

The finite stock-flow statement is

$$\mathbf{p}_i \xrightarrow{\mathbf{J}} \mathbf{p}_f, \quad \boxed{\mathbf{p}_i + \mathbf{J} = \mathbf{p}_f},$$

or

$$\mathbf{J} = \mathbf{p}_f - \mathbf{p}_i = \Delta \mathbf{p}.$$

Here the impulse $\mathbf{J}$ is the momentum flow. Define the average force by

$$\mathbf{F}_{\text{av}} := \frac{\mathbf{J}}{\Delta t},$$

so

$$\mathbf{p}_f - \mathbf{p}_i = \mathbf{J} = \mathbf{F}_{\text{av}} \Delta t.$$

Units:

$$[p] = [J] = \text{kgm/sec} = \text{N sec}, \quad [F] = \text{N}$$

D1. Momentum balance. Write the stock-flow diagram and finite balance for momentum.

D2. Same impulse, different time. Two stopping mechanisms produce the same Δp , but one acts ten times longer. Compare their average-force magnitudes. (Check your seatbelt!)

D3. ★ Instantaneous form. If the local limit exists, explain how the finite balance leads to

$$\mathbf{F} = \frac{d\mathbf{p}}{dt}.$$

Question 5. A 75 kg driver moving at 20 m/sec is brought to rest. Compare the average stopping forces for stopping intervals of 0.20 sec with a seatbelt and 0.020 sec without. Estimate the corresponding masses whose weights equal these force magnitudes, using $|g| = 9.81 \text{ m s}^{-2}$.

Answer.

$$\Delta p = 0 - 75 \text{ kg } 20 \text{ m/sec} = -1500 \text{ kgm/sec}.$$

Hence

$$F_{\text{with}} = \frac{-1500 \text{ kgm/sec}}{0.20 \text{ sec}} = -7.5 \times 10^3 \text{ N}$$

while

$$F_{\text{without}} = \frac{-1500 \text{ kgm/sec}}{0.020 \text{ sec}} = -7.5 \times 10^4 \text{ N}$$

Note that the negative signs show that the average net force on the driver is directed backward. Same impulse, ten times shorter interval, ten times larger average-force magnitude.

The corresponding masses are

$$m_{\text{with}} = \frac{7.5 \times 10^3 \text{ N}}{9.81 \text{ m s}^{-2}} \approx 765 \text{ kg},$$

and

$$m_{\text{without}} = \frac{7.5 \times 10^4 \text{ N}}{9.81 \text{ m s}^{-2}} \approx 7.65 \times 10^3 \text{ kg}.$$

VI. Energy as a Stock

In the simplest work-energy realization,

$$K_i \xrightarrow{\Downarrow W_{\text{net}}} K_f, \quad \boxed{K_i + W_{\text{net}} = K_f},$$

so

$$\boxed{W_{\text{net}} = K_f - K_i = \Delta K}.$$

Here the net work is the flow into the kinetic-energy stock. Define average power by

$$P_{\text{av}} := \frac{W_{\text{net}}}{\Delta t},$$

so

$$\boxed{K_f - K_i = W_{\text{net}} = P_{\text{av}} \Delta t}.$$

Units:

$$[E] = [W] = \text{J}, \quad [P] = \text{J/sec} = \text{W}.$$

E1. Work-energy balance. Identify stock and finite flow; then define average power.

E2. Force and power as rates. Compare

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} \quad \text{and} \quad P = \frac{dE}{dt}.$$

In what structural sense are they analogous?

E3. State versus process. Explain why K_i and K_f label states while W describes a transfer between states.

Question 6. A motor transfers 12 kJ of energy in 4.0 sec. Find the average power. What happens if the same energy transfer takes 12 sec?

Answer. For the first interval,

$$P_{\text{av}} = \frac{12\,000 \text{ J}}{4.0 \text{ sec}} = 3.0 \text{ kW}.$$

For the second,

$$P_{\text{av}} = \frac{12\,000 \text{ J}}{12 \text{ sec}} = 1.0 \text{ kW}.$$

Same energy flow, three times longer interval, one-third the average power.

Mixed Practice and Compact Review

M1. Diagram first. Draw the defining diagram, translate it as $S_i + \mathcal{F} = S_f$, and identify the gain.

M2. One structure, three languages. Write the same finite stock-flow pattern for a cash balance, linear momentum, and kinetic energy.

M3. Conservation diagnostic. In each context, state what must vanish for the initial and final stock to be equal.

M4. Average does not mean constant. Explain why

$$S_f - S_i = \bar{I} \Delta t$$

is exact even when flow occurs in bursts. Give one physical and one financial example.

M5. Momentum comparison. Two objects experience the same impulse. One receives it in 0.50 sec, the other in 0.050 sec. Compare average-force magnitudes.

M6. Energy comparison. A motor transfers 12 kJ in 4.0 sec. Identify stock, finite flow, average rate, and units. What changes if the same transfer occurs in 12 sec?

Master Formulation Sheet

$$S_i \xrightarrow{\Downarrow \mathcal{F}} S_f$$

$$\boxed{S_i + \mathcal{F} = S_f}, \quad \boxed{\mathcal{F} = S_f - S_i}.$$

If time labels are present,

$$\boxed{\mathcal{F} = \bar{I} \Delta t}.$$

Conservation means

$$\boxed{\mathcal{F} = 0 \iff S_f = S_i}.$$

Context	Stock	Flow	Balance
Abstract	S	$\mathcal{F}$	$S_i + \mathcal{F} = S_f$
Momentum	p	J	$p_i + J = p_f$
Energy	K	W	$K_i + W = K_f$
Cash	B	net cash	$B_i + \mathcal{F} = B_f$

★ Instantaneous forms

When the required limits exist,

$$\frac{dS}{dt} = I, \quad F_{\text{force}} = \frac{dp}{dt}, \quad P = \frac{dE}{dt}.$$

These refine rather than replace the finite balances.

Compact master equations

$$\boxed{\mathcal{F} = S_f - S_i = \bar{I} \Delta t},$$

$$\boxed{J = p_f - p_i = F_{\text{av}} \Delta t},$$

$$\boxed{W_{\text{net}} = K_f - K_i = P_{\text{av}} \Delta t}.$$

Quick unit map

Quantity	Units	Meaning
p	kg m/s	momentum stock
J	N s	momentum flow
F	N	momentum-flow rate
E, W	J	energy stock / flow
P	W	energy-flow rate
B	currency	financial stock
$\bar{I}$	stock/time	average flow rate

Five-step answer template

1. System.
2. Stock.
3. Finite flow.
4. Finite balance.
5. Rate, only if useful.

Short-answer prompts

- Why is a finite average rate exact even for a discontinuous history?
- Why is force analogous to power but not to energy?
- Why is impulse analogous to work more closely than to force?
- What is gained by choosing the system before writing an equation?
- What does zero net flow say, and what does it not say, about intermediate behavior?

Finite bookkeeping first; rates second; limits only when necessary.